

RESEARCH STATEMENT

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*When the Model Becomes the Risk: A Research Program in Scalable
Bayesian Inference for Trustworthy Environmental Health Surveillance*

My research builds Bayesian statistical methodology for environmental and health data collected across many locations over many years. The unifying intellectual question of my program is structural: how do we build large hierarchical models that scale to national surveillance data while *preserving the integrity of the scientific conclusions they produce*? I pursue this question through methods rooted in Bayesian statistics, spatial statistics, numerical linear algebra, and dimension reduction, and I develop them in direct partnership with the environmental health applications that motivate them. My goal as an independent investigator is to establish a sustained, externally funded research program — and to train the students who will carry it forward — at the interface of scalable Bayesian methodology and environmental health surveillance.

Research Vision and Significance

Federal and public health agencies build their understanding of environmental risk from data collected across thousands of locations over many years. The validity of every regulatory conclusion drawn from such data depends on the statistical model used to separate an exposure effect from the spatial and temporal background that surrounds it. Hierarchical Bayesian models are the standard tool, but at national scale they fail in a specific and consequential way: combining spatial and temporal random effects produces a large, strongly collinear basis that destabilizes computation and, more seriously, absorbs signal belonging to the exposure of interest — in the worst case reversing its apparent direction, so that a harmful exposure appears protective. When a modeling artifact can flip the sign of a regulatory conclusion, the methodology itself becomes a public health hazard.

The significance of my program lies in treating this not as a nuisance to be patched analysis-by-analysis, but as a foundational problem deserving its own theory. My central contribution, SVD-PICAR, demonstrates that scale and integrity need not be traded against each other: a singular value decomposition of the projected intrinsic conditional autoregressive basis reduces a spatio-temporal model to the few orthogonal directions that genuinely explain the outcome, so the model effectively learns its own dimension rather than relying on analyst judgment. This is a methodological contribution with a clear constituency — NIEHS, CDC-NCHS, NCI, and EPA all confront precisely these models — which positions the program to attract sustained external funding (NIH R01, NSF DMS, and EPA mechanisms) while producing methodology of independent statistical interest.

Foundational Contributions

My dissertation establishes the framework on which my independent program will build. I summarize the two contributions that anchor it.

1. *Compressed spatio-temporal inference for Gaussian and multinomial outcomes.*

Combining spatial and temporal structure produces a basis riddled with near-collinear components that slow computation and destabilize inference. I showed how the optimal low-rank representation can be characterized through the singular value decomposition, and used adversarial simulation designs to establish that the compressed model preserves both accuracy and well-calibrated uncertainty while attaining the speed and stability of a compact model [1]. The method was validated on large real-world datasets, including NASA satellite environmental measurements and national cancer surveillance records.

2. *Confounding-aware mixture effects for over-dispersed count data.*

Bayesian kernel machine regression is the standard tool for estimating the joint effect of correlated chemical exposures, but it does not accommodate joint spatio-temporal dependence in count outcomes; ignoring that dependence biases — and can reverse — the estimated mixture effect. My SVD-PICAR negative-binomial BKMR cleanly separates spatio-temporal background variation from the chemical signal. Applied to county-level thyroid cancer counts and Nebraska pesticide records, it recovered the chemicals genuinely associated with elevated cancer risk — results consistent with established toxicological evidence and unrecoverable using the conventional approach [2]. This foundation also supported a collaborative epidemiologic study linking drought to COPD mortality [3], demonstrating the program's capacity for cross-disciplinary, co-authored science.

An Independent Research Program: Three Funded Thrusts

My program advances along three thrusts, each scoped as an independent line of inquiry suitable to anchor a federal grant proposal and to support graduate student dissertations.

1. *Spatially varying exposure effects (anchor for an NIH R01).* A pesticide's health impact is rarely uniform; it depends on local application intensity, hydrology, and co-exposure profiles. I will embed a spatially varying coefficient process within the compressed basis and derive the SVD-PICAR analogue of restricted spatial regression,

so that the exposure surface is estimated free of spatial confounding by construction. The methodological core — resolving the identifiability contest between a flexible coefficient surface and a flexible confounder surface — is of independent statistical interest, while the deliverable (county-level effect maps with calibrated uncertainty) is directly fundable through environmental health mechanisms. This thrust will support one to two PhD students.

2. *Critical exposure windows (anchor for an NSF DMS proposal)*. Evidence-based exposure limits require knowing not only whether but *when* exposure does harm. I will develop a distributed-lag formulation in which the temporal SVD directions serve a dual role — capturing the background trend and spanning a penalized low-rank lag surface — with a fused penalty enforcing toxicologically plausible smoothness while permitting a sharp window to emerge. The theoretical questions (rank behavior of the joint trend-lag basis; conditions for window identifiability) make this a natural NSF methodology proposal and a strong dissertation problem.
3. *Automatic model selection and reproducible software (broader impact and sustainability)*. The barrier between a sound method and its routine use is the expert judgment the method silently demands. I will develop and theoretically characterize information-criterion and predictive procedures (WAIC, leave-one-out cross-validation, and an SVD marginal-likelihood criterion) that select model rank automatically, establishing consistency guarantees and releasing an open-source pipeline an agency analyst can run without statistical adjudication. The software deliverable is both a broader-impacts component for grant proposals and the mechanism by which the program's methods reach practitioners and accrue citations.

Mentoring, Teaching, and Departmental Synergy

I view a research program and a training program as inseparable. Each thrust above is deliberately scoped to yield well-defined, self-contained dissertation problems, so that students enter at a tractable subproblem and graduate having owned a complete methodological contribution. My experience as a graduate research assistant in the UNMC Water, Climate and Health Program, and as a supplementary instructor and teaching assistant across statistics and mathematics, has shaped a mentoring philosophy built on early ownership, frequent feedback, and pairing every method with the applied problem that gives it meaning. I am equipped to teach core graduate biostatistics — Bayesian methods, hierarchical and mixed models, categorical data, and statistical computing in R and Python — and to develop an advanced elective in spatio-temporal modeling that channels students into the program's thrusts.

My program is designed for synergy. It contributes methodology that collaborators in environmental health, epidemiology, and spatial statistics can adopt directly, and it benefits in turn from access to surveillance data and substantive expertise. My NASA ARSET certifications in satellite-derived environmental data position me to bridge statistical and remote-sensing groups, and my record of co-authored, cross-disciplinary work demonstrates that the program both produces methods of independent interest and embeds them in collaborative science.

Long-Term Trajectory

In the near term, my program produces methodology with an identified constituency and clear funding pathways. Over the longer term, as surveillance systems become larger and more interdependent, I expect questions of robustness to misspecification, graceful degradation, and reproducibility to become as central as scalability, and dimension reduction to remain the organizing principle that makes them tractable. My ambition is to build, at an R1 institution, a durable and externally funded research group recognized for methodology that is statistically rigorous and directly consequential for protecting public health — and for the students it trains to extend that work. I am excited at the prospect of learning, contributing, and making a lasting impact in this growing field.

References

- [1] K. B. Afari, Y. Gwon, "Compressed Spatio-Temporal Basis Functions: An SVD-PICAR Framework for Scalable Gaussian and Multinomial Inference," submitted to *Journal of Computational and Graphical Statistics*, 2026. Under review.
- [2] K. B. Afari, Y. Gwon, "SVD-PICAR Bayesian Kernel Machine Regression Framework for Spatio-Temporal Count Data," submitted to *Statistics in Medicine*, 2026. Under review.
- [3] K. C. Gribben, Y. Gwon, K. Afari, J. Berman, A. Li, B. Fard, et al., "Drought increases the risk of COPD mortality in the United States," submitted to *Environmental Research*, 2026. Under review.